

18	Answer: A
	The wavelength being 5.0 m, the path difference ($147.5 - 135 = 12.5$ m) corresponds to 2.5 wavelengths. Hence, the two waves meet in anti-phase. Intensity = $k(\text{amplitude})^2$, k is a constant $\frac{I}{4I} = \frac{k * A_1^2}{k * A_2^2} \Rightarrow \frac{A_1}{A_2} = \frac{1}{2}, \text{ or } A_2 = 2A_1$ When meet in anti-phase, the two vectors are in opposite direction. The resultant vector is the difference between them: $A_{\text{tot}} = A_2 - A_1 = 2A_1 - A_1 = A_1$ Hence, the resulting intensity is $I_{\text{resultant}} = kA_{\text{tot}}^2 = kA_1^2 = I$

